

HAZBOT

Autonomous Hazardous Material Handling Robot

Mobile Chassis · AI Perception · Robotic Hand · ROS2

3-Layer	ROS2	4-DOF	360°	AI
System Architecture	Navigation Stack	Robotic Hand	LiDAR Coverage	Object Detection

The Problem

Why do we need autonomous hazardous material handling?

3.2M+	340+	$180B	62%
Hazmat incidents reported yearly (US)	Firefighter deaths from hazmat exposure	Annual cost of hazmat accidents	Incidents involve human responders



Chemical Exposure

Workers risk severe chemical burns, poisoning, and long-term respiratory disease when handling unknown substances manually.



Explosive Risk

Mishandling explosive or flammable materials by human operators can cause fatal detonations and infrastructure damage.



Biological Hazards

Pathogens and biological waste require full PPE protocols. Errors in containment can cause pandemic-level outbreaks.

✅ **Solution: HazBot — a fully autonomous robot that keeps humans out of danger zones entirely**

System Architecture Overview

Three integrated layers working as one unified robot



LAYER 1 — MOBILE CHASSIS

ROS2 · Diff-Drive · SLAM · Nav2 · LiDAR



LAYER 2 — PERCEPTION & AI

Depth Camera · YOLO · Hazard Classification



LAYER 3 — ROBOTIC HAND

Servo Motors · 6-DOF · Pick & Place

DATA FLOW



LiDAR Scan



Depth Image



SLAM Map



Nav2 Goal



AI Detection



Pick Command

All layers communicate via ROS2 topics · Real-time processing · Fully autonomous operation

P H A S E 0 1

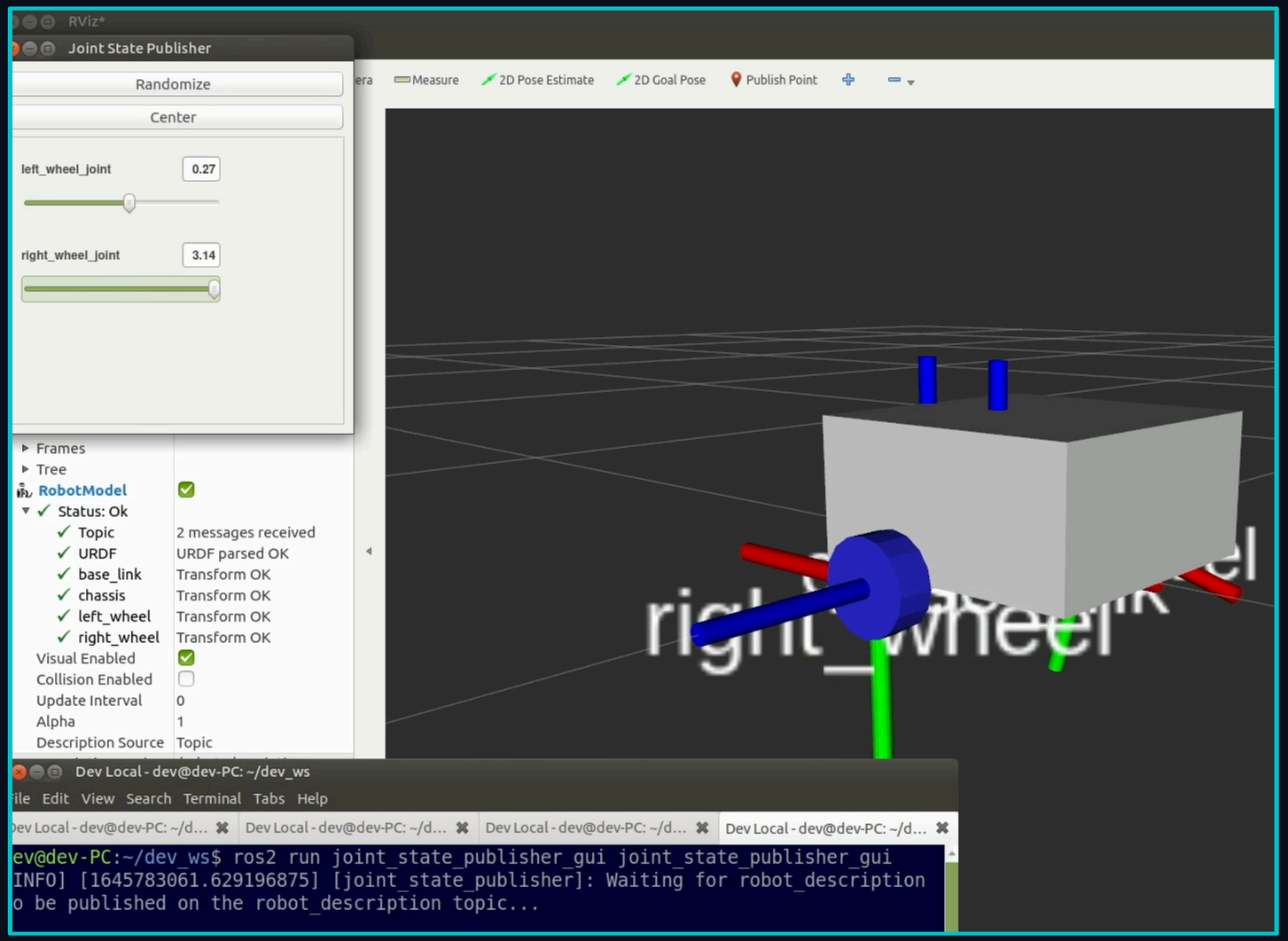


Mobile Chassis

ROS2 · Gazebo Simulation · Differential Drive

Robot Design & Simulation

URDF modeling + Gazebo virtual environment



SIMULATION PIPELINE

1. Write URDF / Xacro
2. Load in Gazebo
3. ros2_control Plugin
4. Test SLAM + Nav2
5. Iterate & Refine
6. **Deploy to Hardware**

Power System Architecture

12V LiPo battery → regulated power for motors, Pi, and sensors

ON DIAGRAM

12V → Motors



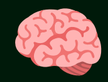
3S LiPo
11.1–12.6V



10A Fuse
+ Switch



Buck Conv.
5V reg.



Raspberry Pi
+ USB Hub

CONNECTED COMPONENTS



L298N
Motor Driver

12V



Arduino Nano
Controller

5V



RPLidar A1
Sensor

5V



OAK-D Lite
Camera

5V



Servo Motors
×5

5V

PHASE 02



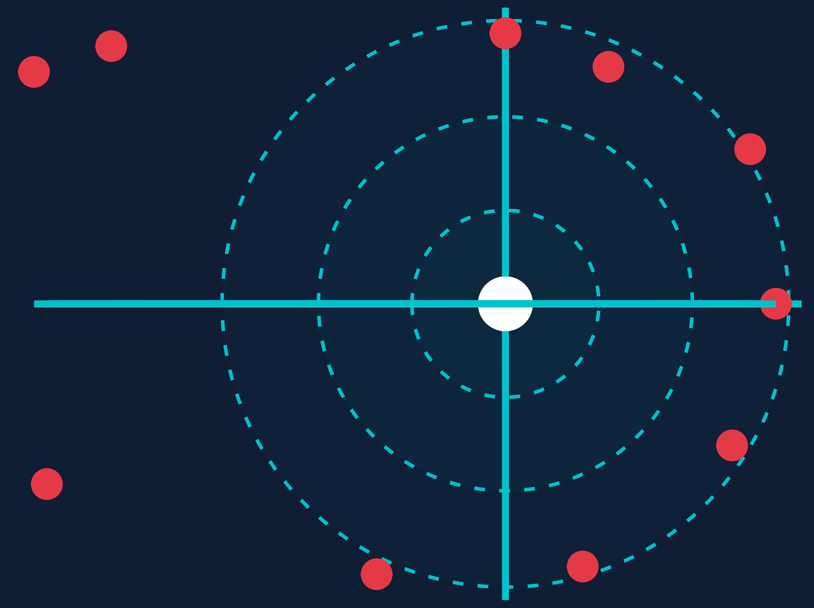
Sensors & Mapping

LiDAR · Depth Camera · SLAM · Nav2

Sensors: LiDAR + Depth Camera

360° environmental awareness + 3D object perception

📡 RPLiDAR A1 — 2D LASER SCAN



360° / 8,000 samples/sec / 12m range

📷 OAK-D LITE — DEPTH PERCEPTION

- **< 0.5m** CLOSE
- **0.5–1m** NEAR
- **1–2m** MID
- **2–4m** FAR
- **> 4m** DISTANT

Baseline	7.5 cm stereo
Depth Range	20cm – 35m
RGB Output	4K @ 30fps
Point Cloud	via depth_image_proc
ROS Topic	/camera/depth/image_raw

SLAM — Simultaneous Localization & Mapping

slam_toolbox builds a live map while tracking robot position

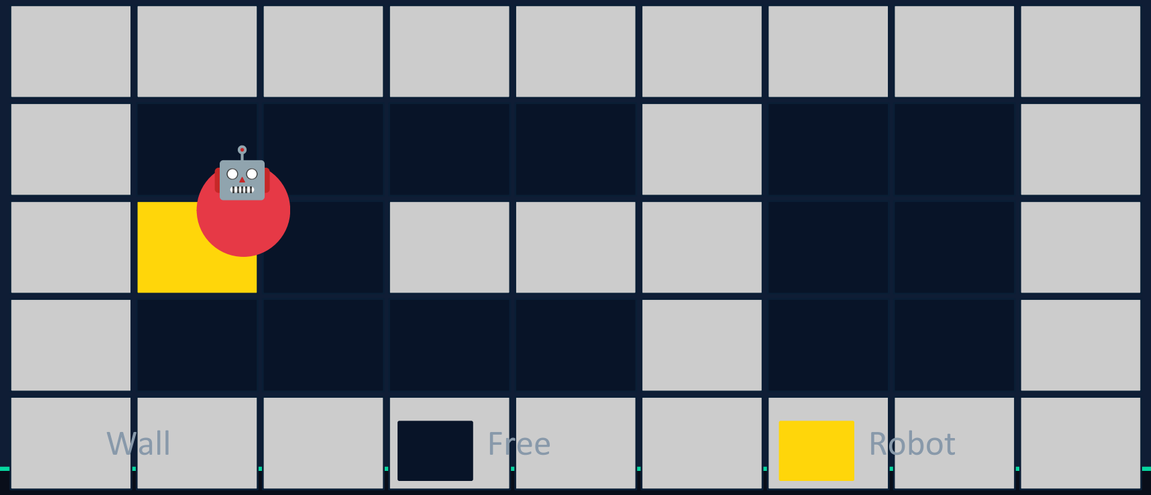
ROS2 FRAME HIERARCHY

map — Global reference

odom — Odometry drift frame

base_link — Robot center

OCCUPANCY GRID MAP



SLAM OPERATIONAL WORKFLOW

1. LiDAR Scan
2. Update Occupancy
3. Estimate Pose
4. Correct Drift
5. Save Map
6. Localize Only

Autonomous Navigation with Nav2

Global planner + local planner + costmaps + twist_mux

NAV2 ARCHITECTURE STACK

2D Goal Pose (RViz)

Behavior Tree (BT Navigator)

Global Planner (A*)

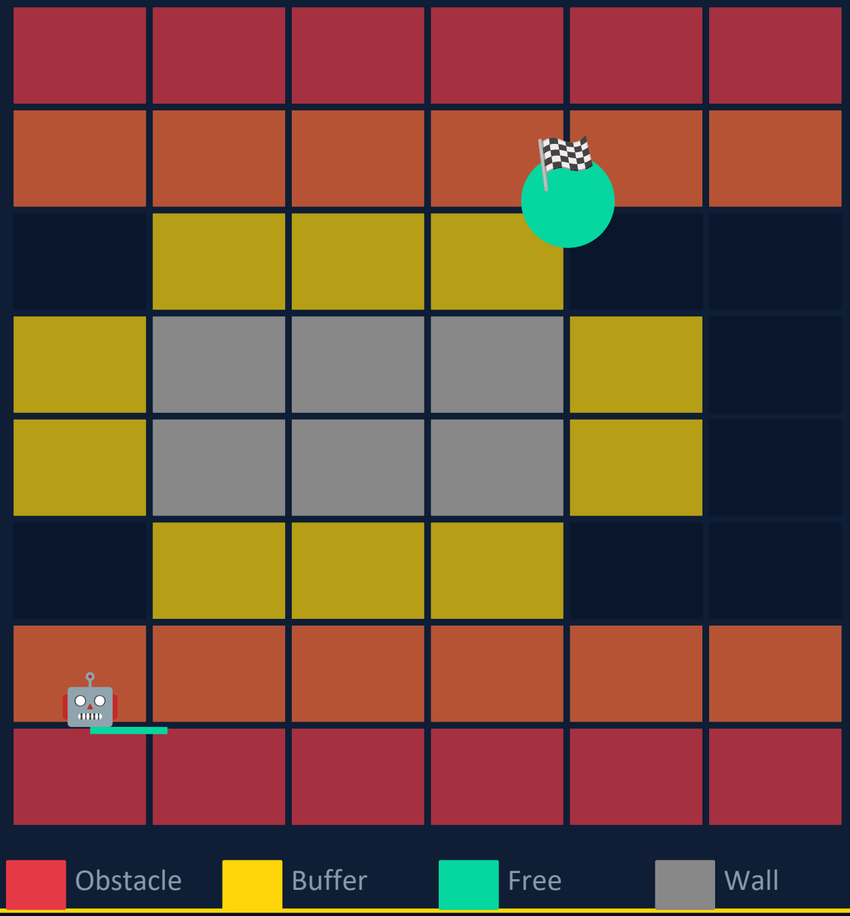
Local Planner (DWA)

Costmap 2D (global + local)

twist_mux → cmd_vel

ros2_control → Motors

COSTMAP VISUALIZATION



PHASE 03



Perception & AI

Object Detection · Hazard Classification · YOLO + Depth Fusion

Object Detection & Identification

YOLO + Depth Camera fusion for real-time hazard detection

AI DETECTION PIPELINE

RGB Frame (OAK-D) → Preprocess Resize → YOLO v8 Inference → Bounding Boxes → Depth Fusion → 3D World Coords → Hazard Alert

SIMULATED DETECTION OUTPUT



DETECTION RESULTS TABLE

Object	Conf	Dist	Risk
Acid Vial	94%	1.2m	🔴 HIGH
Bio-Hazard Bag	87%	2.1m	🟠 MED
Fuel Canister	91%	1.8m	🟡 MED
Safe Container	78%	3.2m	🟢 LOW

Hazard Classification System

Multi-factor risk scoring — detection confidence × hazard type × proximity

RISK MATRIX — CONFIDENCE vs PROXIMITY

	< 1m	1-2m	2-3m	> 3m
HIGH >85%	CRITICAL	CRITICAL	HIGH	MEDIUM
MED 60-85%	CRITICAL	HIGH	MEDIUM	SAFE
LOW <60%	HIGH	MEDIUM	SAFE	SAFE

CRITICAL | HIGH | MEDIUM | SAFE

ROBOT RESPONSE ACTIONS



CRITICAL

Stop immediately
Alert operator
Activate containment



HIGH

Slow approach
Extend hand
Delicate grip



MEDIUM

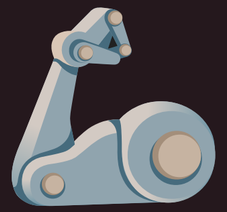
Standard approach
Normal grip force
Log event



SAFE

Normal operation
Pick & transport
No special handling

P H A S E 0 4



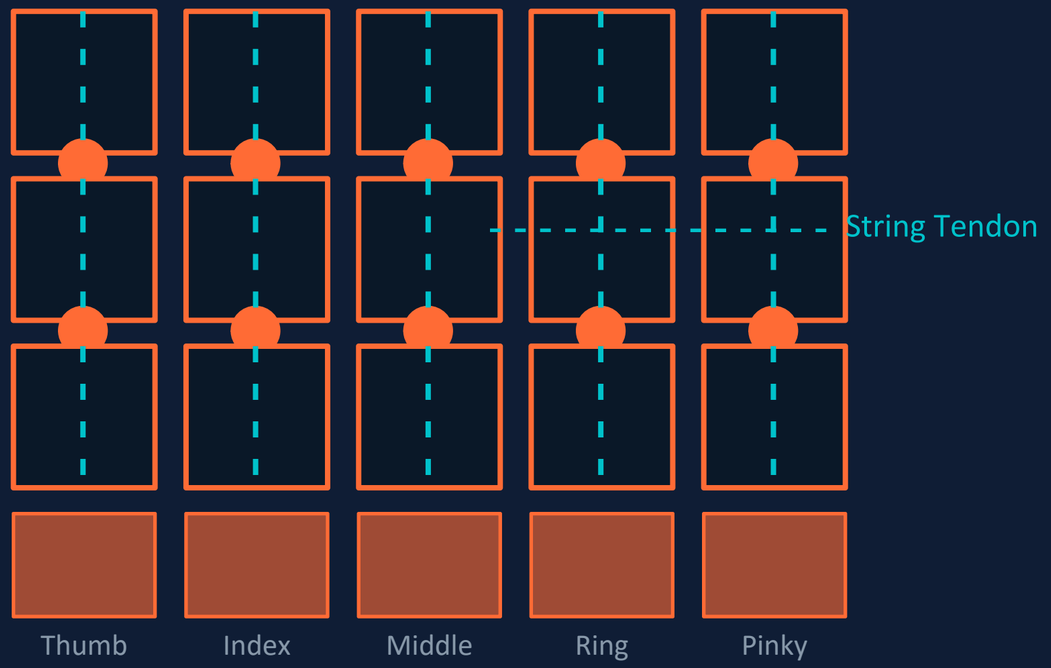
Robotic Hand

Servo-Driven · 6-DOF · Pick & Place Execution

Robotic Hand Mechanism & Integration

Tendon-driven fingers + servo motors mounted on mobile chassis

HAND ANATOMY DIAGRAM



Servo Motors ×5

INTEGRATION SPECIFICATIONS

DOF	6 (5 fingers + wrist rotation)
Actuators	MG996R servo motors × 5
Controller	Arduino Nano via USB serial
Grip Force	~2.5 kg·cm per finger
Payload	Up to 500g per grip
Response	< 200ms command-to-grip
Mount	Pan-tilt bracket on chassis top
Power	5V regulated from buck converter
ROS Interface	custom hand_controller node

Pick & Place Execution Workflow

From hazard detection to safe containment — step by step

1



Detect

LiDAR + AI camera identify hazard object

2



Locate

3D coords via depth-camera fusion

3



Navigate

Nav2 drives robot to 0.5m from target

4



Align

Pan-tilt camera aligns hand precisely

5



Grip

Fingers close on object at correct force

6



Contain

Move to safe zone or containment unit

7



Confirm

Verify placement log event + alert operator

8



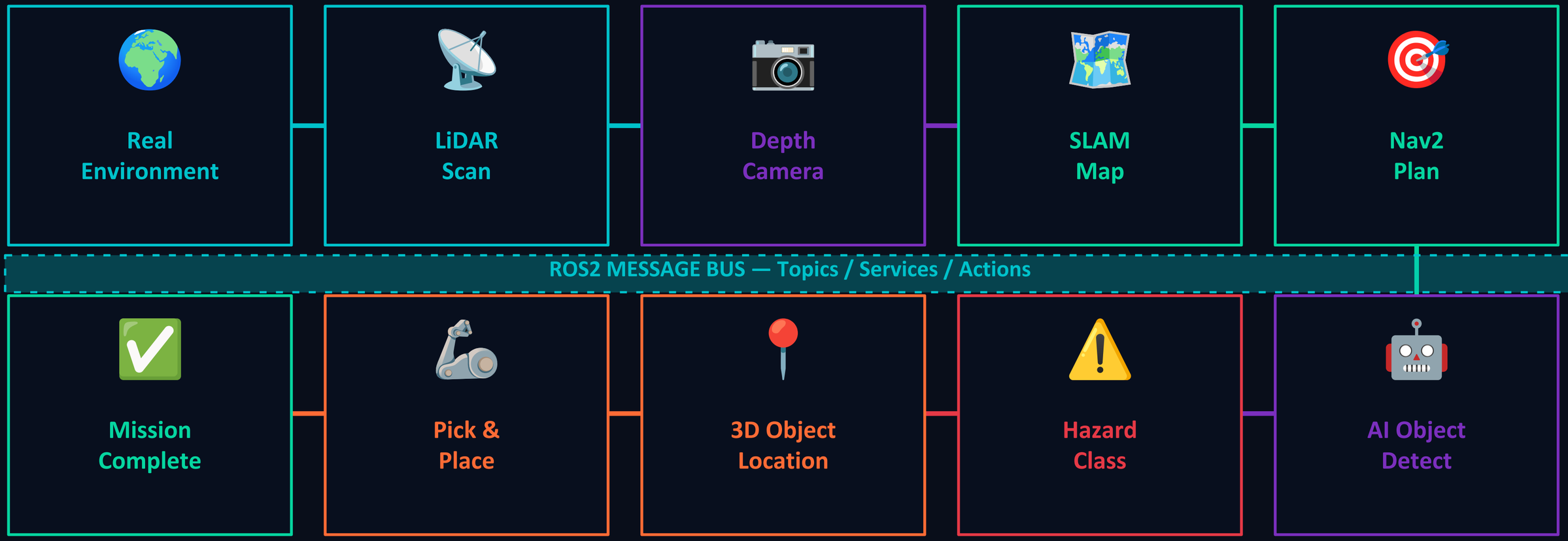
Return

Robot returns to base / next patrol

⏱️ **Total cycle time: ~25–40 seconds per hazmat object | Fully autonomous — zero human contact required**

Full System Integration Flow

Complete data pipeline from environment to action



PERFORMANCE METRICS

< 50ms	< 100ms	360°	< 200ms	98%	10hr
Detection Latency	Nav2 Path Recalc	LiDAR Coverage	Grip Response	Detection Accuracy	Battery Life

Applications, Future Work & Thank You

Real-world impact and the road ahead

Nuclear Plants

Handle radioactive waste in reactor zones

Bio Labs

Manage pathogen samples and biohazard bags

Chemical Factories

Move hazardous chemical containers safely

Fire Response

Clear fuel spills and explosive materials

Mining Sites

Handle toxic minerals and heavy materials

Oil Spill Cleanup

Autonomous containment in dangerous zones

FUTURE IMPROVEMENTS

- AI-powered hazard learning (RL)
- Swarm mode — multiple HazBots
- Wireless charging dock
- Exo-hand with force feedback
- Cloud mission logging & alerts

THANK YOU

HazBot

Questions & Discussion

Keeping humans safe — one pick at a time.